Analysis

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TPOT with graph search space versus ensemble

2000 total evals

Version 1

Stats Testing

```
## # A tibble: 26 x 9
##    task_id .y.   group1            group2    n1    n2 statistic       p p.signif
##      <int> <chr> <chr>             <chr>  <int> <int>     <dbl>   <dbl> <chr>
##  1    2073 auroc ensemble_random_~ indiv~    21    21       205  1   e-3 ***
##  2  146818 auroc ensemble_random_~ indiv~    21    21       176. 8.03e-3 **
##  3  146820 auroc ensemble_random_~ indiv~    21    21       182  2.18e-2 *
##  4  168350 auroc ensemble_random_~ indiv~    21    21       227  6.68e-6 ****
##  5  168757 auroc ensemble_random_~ indiv~    21    21       204  2.22e-3 **
##  6  168784 auroc ensemble_random_~ indiv~    21    21       217  1.05e-4 ***
##  7  168911 auroc ensemble_random_~ indiv~    21    21       202  1.6 e-3 **
##  8  190137 auroc ensemble_random_~ indiv~    21    21       225  1.34e-5 ****
##  9  190146 auroc ensemble_random_~ indiv~    21    21       148  2.72e-1 ns
## 10  190411 auroc ensemble_random_~ indiv~    21    21       193  5.54e-3 **
## 11  359955 auroc ensemble_random_~ indiv~    21    21       170  5.95e-2 ns
## 12  359956 auroc ensemble_random_~ indiv~    21    21       214  1.97e-4 ***
## 13  359958 auroc ensemble_random_~ indiv~    21    21       215  1.61e-4 ***
## 14  359959 auroc ensemble_random_~ indiv~    21    21       225  1.34e-5 ****
## 15  359960 auroc ensemble_random_~ indiv~    21    21        50  2.49e-2 *
## 16  359962 auroc ensemble_random_~ indiv~    21    21       184  1.58e-2 *
## 17  359963 auroc ensemble_random_~ indiv~    21    21       228  4.77e-6 ****
## 18  359964 auroc ensemble_random_~ indiv~    21    21       189  9.02e-3 **
## 19  359965 auroc ensemble_random_~ indiv~    21    21       118  1.63e-1 ns
## 20  359968 auroc ensemble_random_~ indiv~    21    21       149  2.57e-1 ns
## 21  359969 auroc ensemble_random_~ indiv~    21    21       231  9.54e-7 ****
## 22  359970 auroc ensemble_random_~ indiv~    21    21       231  9.54e-7 ****
## 23  359971 auroc ensemble_random_~ indiv~    21    21       212  2.93e-4 ***
## 24  359972 auroc ensemble_random_~ indiv~    21    21       216  1.31e-4 ***
## 25  359974 auroc ensemble_random_~ indiv~    21    21       207  7.21e-4 ***
## 26  359975 auroc ensemble_random_~ indiv~    21    21       122  8.38e-1 ns
```

Medians

```
## # A tibble: 26 x 3
##    task_id individual_tpot ensemble_random_100
##      <int>           <dbl>               <dbl>
##  1    2073           0.881               0.891
##  2  146818           0.936               0.937
##  3  146820           0.995               0.997
##  4  168350           0.965               0.967
##  5  168757           0.791               0.802
##  6  168784           0.963               0.969
##  7  168911           0.868               0.872
##  8  190137           0.919               0.931
##  9  190146           0.967               0.968
## 10  190411           0.921               0.923
## 11  359955           0.743               0.744
## 12  359956           0.933               0.939
## 13  359958           0.942               0.948
## 14  359959           0.742               0.755
## 15  359960           1                   1
## 16  359962           0.837               0.845
## 17  359963           0.995               0.996
## 18  359964           0.994               0.994
## 19  359965           1.00                1.00
```

```
## 20  359968          0.929             0.926
## 21  359969          0.845             0.854
## 22  359970          0.914             0.930
## 23  359971          0.997             0.998
## 24  359972          0.991             0.992
## 25  359974          0.870             0.891
## 26  359975          0.994             0.995
```